APPENDIX 3—CÔTE D'IVOIRE: JOINT BANK-FUND DEBT SUSTAINABILITY ANALYSIS[1]

The debt sustainability analysis (LIC-DSA) shows that Côte d'Ivoire is at a moderate risk of debt distress. After full HIPC, MDRI and beyond-HIPC debt relief, all external debt indicators remain under their indicative thresholds[2] throughout the projection period. However, the country remains vulnerable to certain macroeconomic shocks, especially through the middle of the projection period. The inclusion of domestic debt raises debt burden indicators somewhat, but does not alter the overall assessment.

A. Introduction

1. **This analysis (LIC-DSA) is based on the Debt Sustainability Framework (DSF) for Low-Income Countries**. The DSA presents the projected path of Côte d'Ivoire's external and public sector debt indicators, and assesses the country's risk of external debt distress. The LIC-DSA and the HIPC Initiative Debt Relief Analysis (HIPC-DRA) share the same macroeconomic assumptions for the baseline, but methodologically, they differ. The LIC-DSA compares the evolution over the projection period of debt burden indicators against policy-dependent indicative thresholds. In contrast, under the HIPC-DRA, the historical debt burden indicators are compared to uniform thresholds in order to calculate the amount of HIPC debt relief that Côte d'Ivoire qualifies for in the context of the HIPC Initiative. In addition, the results of the LIC-DSA differ from the HIPC-DRA because of methodological differences related to the definition of the discount rates and the exchange rates used.

2. **The last LIC-DSA[3] for Côte d'Ivoire was prepared in November 2011 and assessed Côte d'Ivoire as being in debt distress**.[4] At that time, some arrears accumulated to a number of creditors during the post-election crisis were still unpaid. The results below are in line with the alternative scenario related to HIPC, MDRI and beyond-HIPC relief in the November 2011 LIC-DSA.

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[1] The DSA was prepared jointly by the staffs of the International Monetary Fund and the World Bank, in collaboration with the authorities of Côte d'Ivoire. The fiscal year in Côte d'Ivoire is January–December.

[2] For the threshold for countries weak policy environment, see http://www.imf.org/external/np/pp/eng/2012/011212.pdf

[3] The last DSA can be found here: http://www.imf.org/external/pubs/cat/longres.aspx?sk=25372.0

[4] The DSAs presented in this document are based on the low-income countries (LIC) DSA framework. Under the Country Policy and Institutional Assessment (CPIA), Côte d'Ivoire is rated as a weak performer with an average rating of 2.72 in 2008–10, and the DSA uses the indicative threshold indicators for countries in this category. See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/020304.htm and IDA/SECM2004/0035, 2/3/04) and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework, Policy Implications" http://www.imf.org/external/np/pdr/sustain/2004/091004.htm and IDA/SECM2004/0629, 9/10/04) and "A Review of Some Aspects of the Low-Income Country Debt Sustainability Framework" (http://www.imf.org/external/np/pp/eng/2009/080509a.pdf) and "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (http://www.imf.org/external/np/pp/eng/2010/012210.pdf).

3. **The dynamic of Côte d'Ivoire's debt has improved because of debt restructuring as discussed in Section III and appendix 1 of this report**.

B. Baseline Assumptions

4. **The assumptions of the LIC-DSA are broadly consistent with projections of the HIPC-DRA (Box 1 above) and those of the November 2011 LIC-DSA**. There are two differences, though. (i) Both this LIC-DSA and the DRA envisage a more ambitious investment path over the medium term, as compared to the last LIC-DSA, reflecting the better than expected performance of the Ivoirien economy in 2011 and early 2012. As a result, the growth path and thus financing needs are also projected to be higher than in the November 2011 LIC-DSA. The new financing needs will be essentially covered by higher commercial borrowing.[5] (ii) The DRA assumed French post-cutoff date ODA claims of €2.3 billion to be cancelled, and the LIC DSA assumes these to be converted in debt-for-development swaps (C2D) over 15 years.

5. **As the impact of the post-election crisis subsides**, **inflation is expected to decline from 4.9 percent in 2011 to 2.5 percent in the medium term and stay below the WAEMU target of 3 percent in the long run**. These projections reflect sound monetary policy at the regional level and efforts by the authorities to curb trading margins for basic goods.

6. **The baseline scenario reflects the full delivery of HIPC completion point**, **MDRI and beyond-HIPC debt relief**, **as in the HIPC-DRA**.

7. **FDI is expected to rise rapidly over the medium term**, and gradually increase over the long term. Following the HIPC completion point, debt service would rise in the medium term before settling at a lower level in the long run. The initial rise in debt service reflects in part the return to more normal levels of debt service payments, including the onset of payments under C2D,[6] the step-up profile of debt service on Eurobonds, and the repayment of arrears, after the exceptional treatment Côte d'Ivoire obtained from the Paris Club in recent years and the accumulation of past debt service arrears to commercial creditors.

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[5] The grant element of the new borrowing turns negative in the long run (Figure 1.a), as the interest on commercial borrowing (8 percent) is higher than the discount rate (4 percent), and the maturity and grace period are shorter (6 years and 1 year, respectively) as compared to concessional borrowing.

[6] Under C2D the existing debt service claims are reprofiled, and when they are repaid they are channeled into development spending.

C. External Debt Sustainability Analysis

Baseline

8. **Under the baseline scenario**, **Côte d'Ivoire's external debt indicators remain below their relevant indicative thresholds throughout the projection period** (Table 1a, Figure 1). The PV of public and publicly guaranteed (PPG) debt relative to GDP declines over the projection period and stays below the threshold. The PV of PPG debt relative to exports also stays below its indicative threshold throughout the projection period, reflecting the projected debt relief and rising exports.

9. **Debt service ratios remain below their indicative thresholds over the projection period**. However, they are projected to rise from their current exceptionally low levels in the medium term, then decline before stabilizing at their long-term levels. Debt service-to-revenue is relatively close to its threshold in the medium-term, indicating that prudent debt management would be crucial, even after the debt relief related to the HIPC completion point. The dynamics of the debt service ratios reflect mainly (i) the end of exceptional debt service relief from Paris Club creditors and the beginning of full debt service payments on remaining debt after the completion point;[7] (ii) the step-up profile of debt service on the Eurobonds, which begins to rise in 2012 for several years; and (iii) repayment of arrears.

Alternative Scenarios and Stress Tests

10. **Côte d'Ivoire's external debt outlook is subject to considerable vulnerabilities**, **especially until the middle of the projection period** (Table 1b, Figure 1). Given the impact of distortions on trend caused by the civil conflict in Côte d'Ivoire, the sensitivity analysis is based on regional averages and standard deviations for all relevant indicators. The sensitivity tests yield high levels of debt and debt service compared to the baseline scenario, and the debt service-to-revenue indicator almost briefly breaches its indicative threshold under the alternative scenario where key variables are set at their historical averages for the entire projection period. They also reveal that Côte d'Ivoire is most vulnerable to a hypothetical

[7]These are €2.3 billion in post-cutoff date ODA debt to France that will be converted in debt-for-development swaps over time, and the €350 million 2011 emergency loan from France.

combination of one-half standard deviation shocks to real GDP growth, exports, GDP deflator, and net non-debt creating flows; this shock would raise the PV of debt-to-GDP, the PV of debt-to-revenue and the debt-service-to revenue ratios relative to their baseline by 6, 31, and 5 percentage points, respectively, by the middle of the projection period. The PV of debt-to-GDP and the debt-service-to revenue indicators breach their indicative thresholds under this shock. The PV of debt-to-exports ratio is most sensitive to less favorable terms on new borrowing and the debt-service-to-exports ratio is most sensitive to lower export growth, lower net non-debt creating flows, and a combination shock.

11. **In light of the results from the baseline and alternative scenarios as well as the stress tests**, **IDA and IMF staffs conclude that Côte d'Ivoire is at a moderate risk of debt distress** (Figure 3). However as the country improves its policy environment in the medium term, the indicative thresholds could be relaxed and the assessment the debt sustainability could improve.

D. Public Sector Debt Sustainability

Baseline

12. **If domestic public debt is included in the analysis**, **Côte d'Ivoire's debt situation deteriorates modestly** (Table 2a, Figure 2). However, public debt ratios would fall over the long run owing to the projected improvement in the macroeconomic outlook. Under the baseline scenario, the PV of total public debt would gradually decline from 65.9 percent of GDP in 2012 to 30.7 percent of GDP at the end of the projection period, with an average of 32.8 percent of GDP. Debt service on total debt would increase throughout the projection period, reflecting largely the external debt service trend described earlier.

Alternative Scenarios and Stress Tests

13. **Public debt dynamics are vulnerable to shocks** (Table 2b, Figure 2).[8] Public debt indicators are most sensitive to the assumptions on GDP growth. If GDP growth were one standard deviation lower than its historical average, the PV of total debt-to-GDP, PV of total debt-to-revenue, and total debt service-to-revenue ratios would be higher than the baseline value by 43, 190 and 20 percentage points, respectively by the middle of the projection period. While the historical value scenarios may not be relevant for Côte d'Ivoire going forward, they point to the need to foster conditions that are conducive to sustained growth.

E. Conclusion

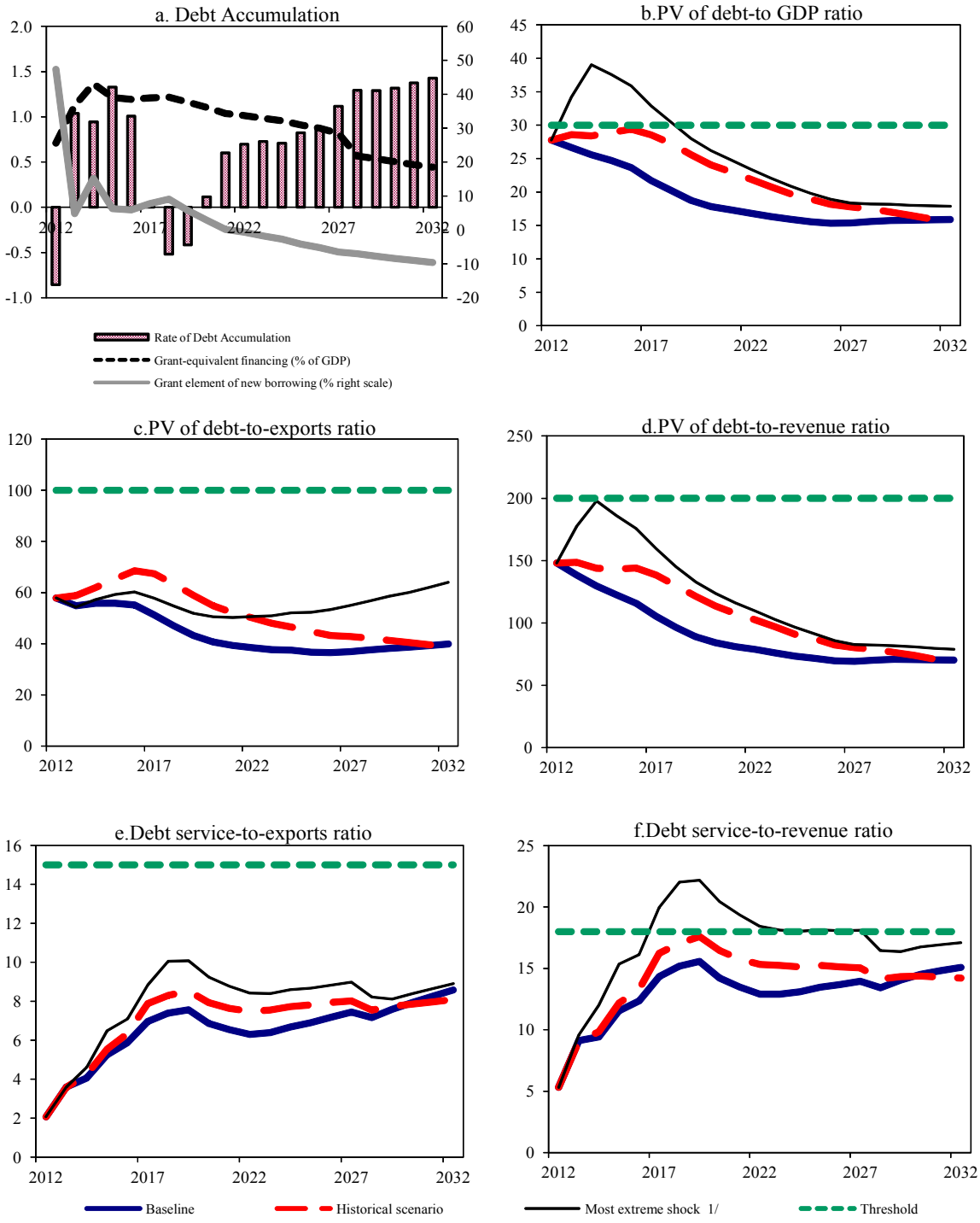
14. **This LIC-DSA shows that Côte d'Ivoire would be at moderate risk of debt distress after the HIPC completion point**. This compares to the "in debt distress"

8 The stress test related to fixed primary fiscal balance used the 2012 value; the 2011 value could not be used, as that year was affected by many temporary factors (post-election crisis and historically high cocoa prices in particular).

assessment in the November 2011 LIC-DSA. Under the current baseline scenario the HIPC Initiative, MDRI and beyond-HIPC debt relief significantly improve the external debt indicators over the projection period. All debt burden indicators remain below their policy-dependent thresholds throughout the projection period, notwithstanding the initial rise in the debt service ratios before they level off in the long run. Alternative scenarios and bound tests reveal the vulnerability of Côte d'Ivoire's external debt outlook, as most indicators rise close to their thresholds or breach them under these scenarios. The inclusion of domestic debt moderately weakens the debt outlook, but does not alter the assessment of Côte d'Ivoire's risk of debt distress.

15. **A sustainable external debt position can be maintained through sound macroeconomic policies and prudent debt management**. Debt relief under the HIPC Initiative, MDRI and beyond-HIPC assistance significantly provides space for Côte d'Ivoire to access some non-concessional financing, increasing its ability to address its significant public investment needs and improving its growth prospects. However, sustained growth, solid export and fiscal revenue performance and prudent debt management are important for maintaining debt sustainability. This is important as the debt service-to-revenue ratio is expected to increase rapidly and be relatively close to its indicative threshold in the medium term. In addition, the rise in debt service-to-revenue ratio underscores the need to improve revenue mobilization and to reduce the vulnerability of revenue to commodities prices through export diversification. The alternative scenarios and stress tests highlight further these conclusions, and the need to avoid unfavorable debt positions.

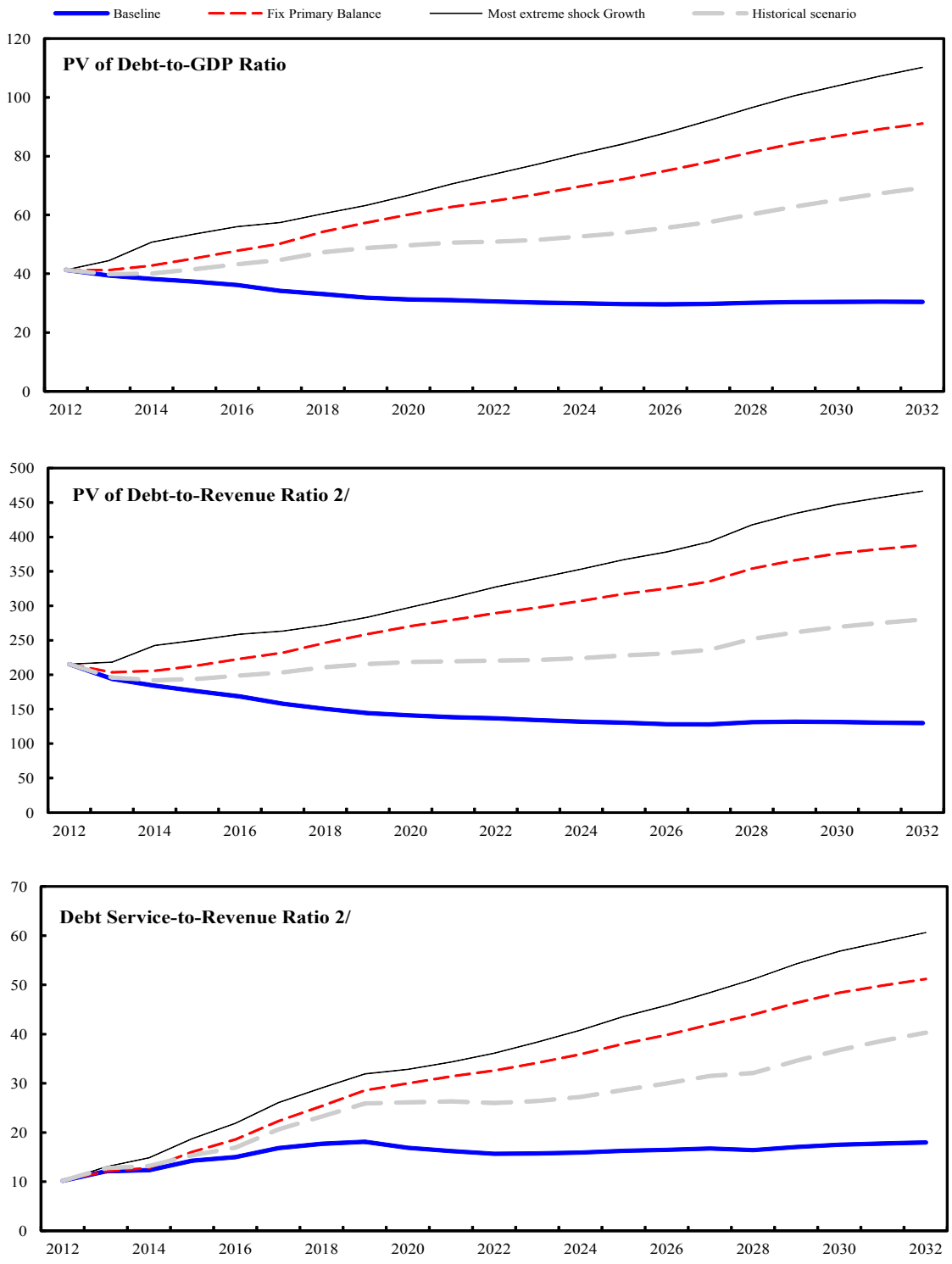
Figure 1. Côte d'Ivoire: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Combination shock; in c. to a Terms shock; in d. to a Combination shock; in e. to a Non-debt flows shock and in figure f. to a Combination shock

Figure 2. Côte d'Ivoire: Indicators of Public Debt Under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Table 1a. Côte d'Ivoire: External Debt Sustainability Framework, Baseline Scenario, 2009-2032 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2009	2010	2011	0 0		2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) 1/	**81.2**	**76.7**	**79.3**			**50.0**	**46.8**	**44.1**	**41.6**	**38.9**	**35.6**		**26.4**	**18.6**	
o/w public and publicly guaranteed (PPG)	53.9	50.6	54.7			26.9	25.8	25.0	24.3	23.3	21.6		17.3	16.1	
Change in external debt	-8.2	-4.6	2.6			-29.3	-3.2	-2.7	-2.5	-2.7	-3.3		-1.4	-0.5	
Identified net debt-creating flows	-4.4	-4.2	-11.3			-5.4	-2.5	-1.8	-2.0	-2.2	-1.7		-0.2	0.5	
Non-interest current account deficit	**-9.0**	**-2.8**	**-8.8**	**-6.0**	**2.8**	**1.3**	**1.2**	**2.5**	**2.9**	**3.5**	**3.9**		**5.2**	**6.0**	5.3
Deficit in balance of goods and services	-11.8	-7.3	-12.2			-3.6	-3.1	-1.6	-1.0	-0.4	-0.3		-1.2	-5.6	
Exports	50.9	51.2	51.2			48.0	48.6	45.8	44.3	42.9	42.4		43.7	39.8	
Imports	39.0	43.9	39.0			44.4	45.5	44.2	43.2	42.5	42.0		42.5	34.2	
Net current transfers (negative = inflow)	0.4	1.8	1.2	2.3	1.1	2.3	2.2	2.1	2.0	2.0	2.3		4.4	10.3	6.0
o/w official	-2.2	-0.8	-1.4			-0.2	-0.2	-0.2	-0.2	-0.2	-0.2		-0.2	-0.2	
Other current account flows (negative = net inflow)	2.4	2.6	2.2			2.6	2.0	2.0	2.0	2.0	1.9		2.0	1.3	
Net FDI (negative = inflow)	**-1.7**	**-1.5**	**-1.1**	**-1.7**	**0.3**	**-2.2**	**-2.8**	**-3.4**	**-4.1**	**-4.9**	**-4.9**		**-5.4**	**-5.6**	-5.5
Endogenous debt dynamics 2/	**6.2**	**0.1**	**-1.3**			**-4.4**	**-0.8**	**-0.9**	**-0.8**	**-0.9**	**-0.7**		**0.0**	**0.1**	
Contribution from nominal interest rate	2.2	1.8	2.3			1.7	2.1	1.9	1.8	1.7	1.7		1.3	1.0	
Contribution from real GDP growth	-3.5	-1.9	3.5			-6.2	-2.9	-2.8	-2.7	-2.6	-2.4		-1.3	-0.9	
Contribution from price and exchange rate changes	7.5	0.3	-7.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.7**	**-0.4**	**13.9**			**-23.9**	**-0.7**	**-0.9**	**-0.5**	**-0.5**	**-1.6**		**-1.3**	**-1.0**	
o/w exceptional financing	-5.3	-2.6	-0.3			-21.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	75.9			50.8	47.6	44.7	42.0	39.3	35.7		25.9	18.4	
In percent of exports	...	...	148.3			105.9	98.0	97.6	95.0	91.5	84.3		59.3	46.2	
PV of PPG external debt	**...**	**...**	**51.4**			**27.8**	**26.7**	**25.6**	**24.7**	**23.7**	**21.7**		**16.8**	**15.9**	
In percent of exports	**...**	**...**	**100.4**			**57.9**	**54.9**	**55.8**	**55.9**	**55.1**	**51.3**		**38.4**	**39.9**	
In percent of government revenues	**...**	**...**	**344.6**			**148.1**	**138.6**	**129.7**	**122.5**	**115.8**	**105.5**		**78.7**	**70.2**	
Debt service-to-exports ratio (in percent)	**-4.0**	**-0.2**	**5.8**			**1.3**	**2.8**	**3.6**	**4.5**	**4.5**	**5.5**		**4.5**	**6.4**	
PPG debt service-to-exports ratio (in percent)	**4.8**	**3.6**	**5.7**			**2.1**	**3.6**	**4.1**	**5.3**	**5.9**	**7.0**		**6.3**	**8.6**	
PPG debt service-to-revenue ratio (in percent)	**13.0**	**9.6**	**19.5**			**5.3**	**9.1**	**9.4**	**11.5**	**12.4**	**14.3**		**12.9**	**15.1**	
Total gross financing need (Billions of U.S. dollars)	-1.8	0.0	-0.7			0.9	0.9	1.2	1.2	1.1	1.4		1.7	3.5	
Non-interest current account deficit that stabilizes debt ratio	-0.8	1.8	-11.4			30.5	4.4	5.2	5.4	6.2	7.3		6.7	6.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.7	2.4	-4.7	0.6	2.5	8.1	6.2	6.5	6.6	6.7	6.7	6.8	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	-7.8	-0.3	10.1	8.3	8.3	-3.2	2.0	1.4	1.9	1.9	2.1	1.0	2.1	3.4	1.9
Effective interest rate (percent) 5/	2.4	2.3	3.1	3.4	1.1	2.3	4.4	4.4	4.6	4.5	4.6	4.1	5.0	5.8	5.2
Growth of exports of G&S (US dollar terms, in percent)	-0.1	2.6	4.9	11.2	9.7	-1.8	9.7	1.8	5.0	5.4	7.5	4.6	6.3	7.1	6.6
Growth of imports of G&S (US dollar terms, in percent)	-9.4	14.8	-6.8	10.9	12.1	19.0	11.1	5.0	6.1	6.9	7.8	9.3	5.6	5.5	5.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47.4	4.8	15.2	6.3	5.9	7.8	14.6	-0.7	-9.6	-3.0
Government revenues (excluding grants, in percent of GDP)	18.9	19.2	14.9			18.7	19.2	19.7	20.2	20.4	20.6		21.4	22.7	21.8
Aid flows (in Billions of US dollars) 6/	0.1	0.1	0.1			0.2	0.4	0.5	0.6	0.6	0.6		0.8	1.1	
o/w Grants	0.1	0.1	0.1			0.1	0.3	0.3	0.3	0.4	0.4		0.5	0.8	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.1	0.2	0.2	0.2	0.2		0.2	0.2	
Grant-equivalent financing (in percent of GDP) 7/	...	...	...			0.7	1.1	1.4	1.2	1.2	1.2		1.0	0.4	0.8
Grant-equivalent financing (in percent of external financing) 7/	...	...	...			68.4	38.5	42.5	31.0	32.0	36.7		26.9	9.7	22.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	22.5	23.0	24.1			25.2	27.3	29.5	32.0	34.8	38.0		50.7	104.8	
Nominal dollar GDP growth	-4.3	2.1	4.9			4.6	8.3	8.0	8.6	8.8	8.9	7.9	7.3	8.6	7.0
PV of PPG external debt (in Billions of US dollars)			11.7			7.0	7.3	7.5	7.9	8.2	8.2		8.5	16.7	
(PVt-PVt-1)/GDPt-1 (in percent)						-19.5	1.0	0.9	1.3	1.0	0.0	-2.5	0.7	1.4	0.8
Gross remittances (Billions of US dollars)	-0.6	-0.6	-0.6			-0.6	-0.7	-0.7	-0.7	-0.7	-0.9		-2.3	-11.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	52.7			28.5	27.3	26.2	25.3	24.2	22.3		17.6	17.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	105.6			61.1	57.7	58.7	58.7	58.0	54.4		43.0	54.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	6.0			2.2	3.8	4.3	5.5	6.2	7.4		7.0	11.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Defined as grants, concessional loans, and debt relief.

7/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	28	27	26	25	24	22	**17**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	28	29	28	29	29	29	**22**	15
A2. New public sector loans on less favorable terms in 2012-2032 2	28	26	26	26	26	24	**22**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	28	28	28	27	25	23	**17**	17
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	28	31	33	32	31	28	**20**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	28	28	28	27	26	24	**18**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	28	32	37	35	34	31	**22**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	28	34	39	38	36	33	**23**	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	28	37	36	34	33	30	**22**	22
PV of debt-to-exports ratio								
Baseline	58	55	56	56	55	51	**38**	40
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	58	59	62	65	69	67	**50**	39
A2. New public sector loans on less favorable terms in 2012-2032 2	58	54	57	59	60	58	**51**	64
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	58	54	55	55	54	50	**36**	38
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	58	68	79	78	77	72	**50**	44
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	58	54	55	55	54	50	**36**	38
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	58	66	80	80	79	73	**50**	41
B5. Combination of B1-B4 using one-half standard deviation shocks	58	69	76	76	75	70	**48**	40
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	58	54	55	55	54	50	**36**	38
PV of debt-to-revenue ratio								
Baseline	148	139	130	123	116	106	**79**	70
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	148	149	144	143	144	139	**103**	68
A2. New public sector loans on less favorable terms in 2012-2032 2	148	137	133	130	127	119	**104**	112
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	148	144	140	131	124	112	**81**	74
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	148	159	169	159	150	136	**94**	71
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	148	146	143	135	127	115	**83**	76
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	148	166	186	175	165	151	**103**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	148	178	198	186	176	160	**110**	79
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	148	195	181	170	160	145	**105**	95

Table 1b. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	2	4	4	5	6	7	**6**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	2	4	4	6	6	8	**7**	8
A2. New public sector loans on less favorable terms in 2012-2032 2	2	4	4	5	5	6	**5**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	2	4	4	5	6	7	**6**	8
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	2	4	5	7	7	9	**8**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	2	4	4	5	6	7	**6**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	2	4	5	6	7	9	**8**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	2	4	5	6	7	9	**8**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	2	4	4	5	6	7	**6**	8

Debt service-to-revenue ratio

Baseline	5	9	9	12	12	14	**13**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	5	9	10	12	13	16	**15**	14
A2. New public sector loans on less favorable terms in 2012-2032 2	5	9	9	10	11	12	**11**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	5	10	10	13	13	15	**14**	16
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	5	9	10	13	14	17	**16**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	5	10	11	13	14	16	**14**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	5	9	11	14	15	18	**17**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	5	10	12	15	16	20	**18**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	5	13	13	16	17	20	**18**	21
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-15	-15	-15	-15	-15	-15	**-15**	-15

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Côte d'Ivoire: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average	Standard Deviation	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	66.5	66.4	69.3			40.5	38.5	37.6	36.8	35.8	34.0		31.1	30.7	
o/w foreign-currency denominated	53.9	50.6	54.7			26.9	25.8	25.0	24.3	23.3	21.6		17.3	16.1	
Change in public sector debt	-8.8	0.0	2.8			-28.8	-1.9	-0.9	-0.8	-1.0	-1.8		-0.3	-0.1	
Identified debt-creating flows	-7.6	-1.0	4.2			-23.6	0.2	-0.4	-1.3	-1.6	-2.2		0.2	-0.2	
Primary deficit	0.6	1.3	2.5	0.3	1.1	3.3	1.7	0.9	0.2	-0.1	-0.8	0.9	0.8	0.5	0.3
Revenue and grants	19.5	19.7	15.2			19.2	20.3	20.8	21.2	21.5	21.6		22.4	23.5	
of which: grants	0.6	0.5	0.3			0.4	1.0	1.0	1.0	1.0	1.0		1.0	0.8	
Primary (noninterest) expenditure	20.1	21.0	17.7			22.5	21.9	21.7	21.4	21.3	20.8		23.2	24.0	
Automatic debt dynamics	-2.9	0.4	1.9			-5.6	-1.5	-1.4	-1.5	-1.5	-1.4		-0.6	-0.8	
Contribution from interest rate/growth differential	-3.4	-1.9	3.8			-4.9	-1.4	-1.4	-1.4	-1.4	-1.4		-0.2	-0.2	
of which: contribution from average real interest rate	-0.7	-0.4	0.5			0.3	1.0	1.0	1.0	0.9	0.9		1.3	1.2	
of which: contribution from real GDP growth	-2.7	-1.6	3.3			-5.2	-2.4	-2.4	-2.3	-2.3	-2.2		-1.5	-1.5	
Contribution from real exchange rate depreciation	0.5	2.3	-1.9			-0.7	-0.1	0.0	-0.1	0.0	-0.1		...	...	
Other identified debt-creating flows	-5.3	-2.6	-0.3			-21.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-5.3	-2.6	-0.3			-21.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.2	0.9	-1.3			-5.2	-2.1	-0.5	0.5	0.6	0.4		-0.5	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	65.9			41.3	39.4	38.2	37.3	36.2	34.2		30.6	30.5	
o/w foreign-currency denominated	...	...	51.4			27.8	26.7	25.6	24.7	23.7	21.7		16.8	15.9	
o/w external	...	...	51.4			27.8	26.7	25.6	24.7	23.7	21.7		16.8	15.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.6	4.2	6.8			5.3	4.1	3.5	3.2	3.1	2.8		4.3	4.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	433.6			215.4	194.1	184.1	175.9	168.6	158.1		136.7	129.8	
PV of public sector debt-to-revenue ratio (in percent)	…	…	442.0			220.2	204.5	193.7	184.9	177.2	166.0		143.3	134.4	
o/w external 3/	…	…	344.6			148.1	138.6	129.7	122.5	115.8	105.5		78.7	70.2	
Debt service-to-revenue and grants ratio (in percent) 4/	15.6	14.6	28.0			10.2	12.1	12.3	14.2	15.0	16.8		15.7	18.0	
Debt service-to-revenue ratio (in percent) 4/	16.1	15.0	28.5			10.4	12.7	13.0	15.0	15.7	17.7		16.4	18.6	
Primary deficit that stabilizes the debt-to-GDP ratio	9.4	1.4	-0.3			32.1	3.6	1.9	1.0	0.9	1.0		1.2	0.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.7	2.4	-4.7	0.6	2.5	8.1	6.2	6.5	6.6	6.7	6.7	6.8	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	1.0	0.8	1.9	1.8	0.5	0.9	3.4	3.5	3.7	3.7	3.9	3.2	4.6	5.9	5.0
Average real interest rate on domestic debt (in percent)	3.0	2.9	0.0	0.7	1.8	2.5	3.6	4.1	4.0	4.0	3.8	3.7	3.7	2.5	4.0
Real exchange rate depreciation (in percent, + indicates depreciation)	0.9	4.4	-3.6	-1.5	7.9	-1.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.0	1.9	5.0	3.3	2.4	2.4	1.9	1.7	1.9	1.9	2.1	2.0	2.1	3.4	1.9
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.1	-0.2	0.0	0.1	0.4	0.0	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	47.4	4.8	15.2	6.3	5.9	7.8	14.6	-0.7	-9.6	...

Sources: Country authorities; and staff estimates and projections.
1/ The public sector includes the central government and select public enterprises.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 2b. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public Debt 2012-2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	41	39	38	37	36	34	31	30
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	41	40	40	42	43	45	51	69
A2. Primary balance is unchanged from 2012	41	41	43	45	48	50	65	91
A3. Permanently lower GDP growth 1/	41	40	39	39	38	37	39	61
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	41	44	51	54	56	57	74	110
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	41	39	38	37	36	34	31	31
B3. Combination of B1-B2 using one half standard deviation shocks	41	41	42	45	47	48	61	92
B4. One-time 30 percent real depreciation in 2013	41	51	50	48	47	45	42	48
B5. 10 percent of GDP increase in other debt-creating flows in 2013	41	51	49	48	47	44	41	40
PV of Debt-to-Revenue Ratio 2/								
Baseline	215	194	184	176	169	158	137	130
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	215	196	192	194	199	204	221	280
A2. Primary balance is unchanged from 2012	215	203	206	213	223	232	289	388
A3. Permanently lower GDP growth 1/	215	196	188	182	177	170	174	257
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	215	218	242	250	259	263	327	467
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	215	192	185	177	169	159	137	130
B3. Combination of B1-B2 using one half standard deviation shocks	215	201	203	209	216	219	270	388
B4. One-time 30 percent real depreciation in 2013	215	252	239	227	218	206	189	203
B5. 10 percent of GDP increase in other debt-creating flows in 2013	215	250	237	227	218	205	182	169
Debt Service-to-Revenue Ratio 2/								
Baseline	10	12	12	14	15	17	16	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	13	13	15	17	21	26	40
A2. Primary balance is unchanged from 2012	10	12	13	16	19	22	33	51
A3. Permanently lower GDP growth 1/	10	12	12	15	16	18	19	33
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	10	13	15	19	22	26	36	61
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	10	12	12	14	15	17	16	18
B3. Combination of B1-B2 using one half standard deviation shocks	10	13	14	15	17	21	29	50
B4. One-time 30 percent real depreciation in 2013	10	14	16	19	21	24	27	37
B5. 10 percent of GDP increase in other debt-creating flows in 2013	10	12	15	21	21	22	22	24

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.